

Thermodynamics

Instructor

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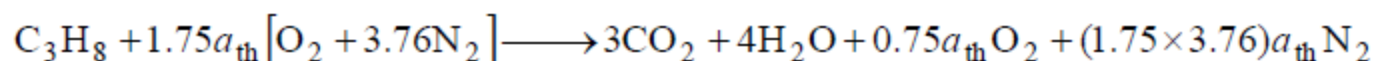
Chapter 15 – Problems

**Problem numbers correspond to
5th Edition**

Theoretical and Actual Combustion Processes

15-14 Propane (C_3H_8) is burned with 75 percent excess air during a combustion process. Assuming complete combustion, determine the air-fuel ratio. *Answer: 27.5 kg air/kg fuel*

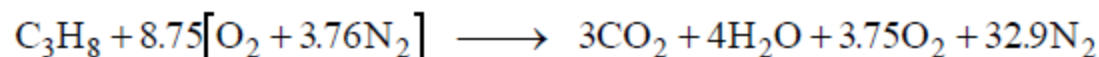
Analysis The combustion equation in this case can be written as



where a_{th} is the stoichiometric coefficient for air. We have automatically accounted for the 75% excess air by using the factor $1.75a_{th}$ instead of a_{th} for air. The stoichiometric amount of oxygen ($a_{th}O_2$) will be used to oxidize the fuel, and the remaining excess amount ($0.75a_{th}O_2$) will appear in the products as free oxygen. The coefficient a_{th} is determined from the O_2 balance,

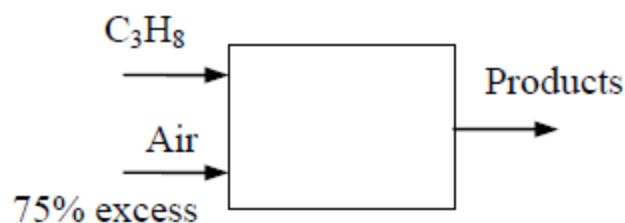
$$O_2 \text{ balance: } 1.75a_{th} = 3 + 2 + 0.75a_{th} \longrightarrow a_{th} = 5$$

Substituting,



The air-fuel ratio is determined by taking the ratio of the mass of the air to the mass of the fuel,

$$AF = \frac{m_{air}}{m_{fuel}} = \frac{(8.75 \times 4.76 \text{ kmol})(29 \text{ kg/kmol})}{(3 \text{ kmol})(12 \text{ kg/kmol}) + (4 \text{ kmol})(2 \text{ kg/kmol})} = 27.5 \text{ kg air/kg fuel}$$



15-15 Acetylene (C_2H_2) is burned with stoichiometric amount of air during a combustion process. Assuming complete combustion, determine the air-fuel ratio on a mass and on a mole basis.

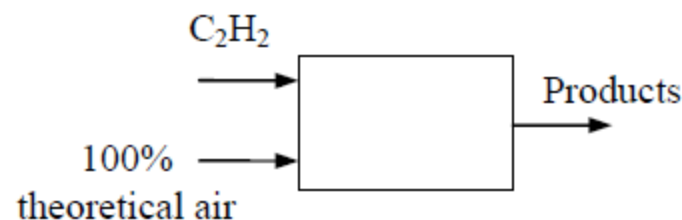
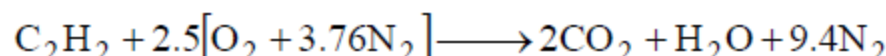
15–15 Acetylene (C_2H_2) is burned with stoichiometric amount of air during a combustion process. Assuming complete combustion, determine the air–fuel ratio on a mass and on a mole basis.

Analysis This is a theoretical combustion process since C_2H_2 is burned completely with stoichiometric amount of air. The stoichiometric combustion equation of C_2H_2 is



$$O_2 \text{ balance:} \quad a_{th} = 2 + 0.5 \quad \longrightarrow \quad a_{th} = 2.5$$

Substituting,



The air–fuel ratio is determined by taking the ratio of the mass of the air to the mass of the fuel,

$$AF = \frac{m_{air}}{m_{fuel}} = \frac{(2.5 \times 4.76 \text{ kmol})(29 \text{ kg/kmol})}{(2 \text{ kmol})(12 \text{ kg/kmol}) + (1 \text{ kmol})(2 \text{ kg/kmol})} = \mathbf{13.3 \text{ kg air/kg fuel}}$$

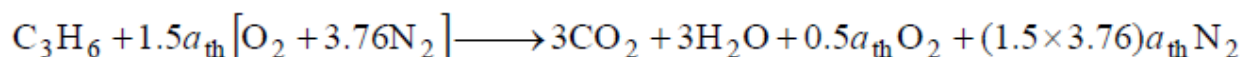
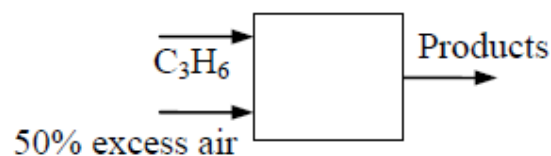
On a mole basis, the air–fuel ratio is expressed as the ratio of the mole numbers of the air to the mole numbers of the fuel,

$$AF_{\text{mole basis}} = \frac{N_{air}}{N_{fuel}} = \frac{(2.5 \times 4.76) \text{ kmol}}{1 \text{ kmol fuel}} = \mathbf{11.9 \text{ kmol air/kmol fuel}}$$

15–18 Propylene (C_3H_6) is burned with 50 percent excess air during a combustion process. Assuming complete combustion and a total pressure of 105 kPa, determine (a) the air–fuel ratio and (b) the temperature at which the water vapor in the products will start condensing.

Properties The molar masses of C, H_2 , and air are 12 kg/kmol, 2 kg/kmol, and 29 kg/kmol, respectively (Table A-1).

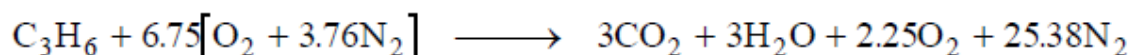
Analysis (a) The combustion equation in this case can be written as



where a_{th} is the stoichiometric coefficient for air. It is determined from

$$O_2 \text{ balance: } 1.5a_{th} = 3 + 1.5 + 0.5a_{th} \longrightarrow a_{th} = 4.5$$

Substituting,



The air–fuel ratio is determined by taking the ratio of the mass of the air to the mass of the fuel,

$$AF = \frac{m_{air}}{m_{fuel}} = \frac{(6.75 \times 4.76 \text{ kmol})(29 \text{ kg/kmol})}{(3 \text{ kmol})(12 \text{ kg/kmol}) + (3 \text{ kmol})(2 \text{ kg/kmol})} = 22.2 \text{ kg air/kg fuel}$$

(b) The dew-point temperature of a gas–vapor mixture is the saturation temperature of the water vapor in the product gases corresponding to its partial pressure. That is,

$$P_v = \left(\frac{N_v}{N_{prod}} \right) P_{prod} = \left(\frac{3 \text{ kmol}}{33.63 \text{ kmol}} \right) (105 \text{ kPa}) = 9.367 \text{ kPa}$$

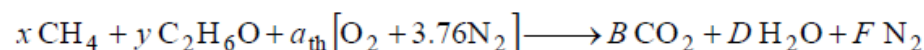
Thus,

$$T_{dp} = T_{sat@9.367 \text{ kPa}} = 44.5^\circ\text{C}$$

15–20 Butane (C_4H_{10}) is burned in 200 percent theoretical air. For complete combustion, how many kmol of water must be sprayed into the combustion chamber per kmol of fuel if the products of combustion are to have a dew-point temperature of 60°C when the product pressure is 100 kPa?

15-21 A fuel mixture of 20 percent by mass methane (CH_4) and 80 percent by mass ethanol ($\text{C}_2\text{H}_6\text{O}$), is burned completely with theoretical air. If the total flow rate of the fuel is 31 kg/s, determine the required flow rate of air. **Answer: 330 kg/s**

Analysis The combustion equation in this case can be written as



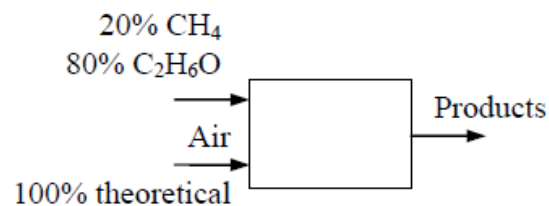
where a_{th} is the stoichiometric coefficient for air. The coefficient a_{th} and other coefficients are to be determined from the mass balances

$$\text{Carbon balance:} \quad x + 2y = B$$

$$\text{Hydrogen balance:} \quad 4x + 6y = 2D$$

$$\text{Oxygen balance:} \quad 2a_{\text{th}} + y = 2B + D$$

$$\text{Nitrogen balance:} \quad 3.76a_{\text{th}} = F$$



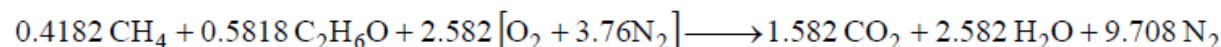
Solving the above equations, we find the coefficients as

$$x = 0.4182 \quad B = 1.582$$

$$y = 0.5818 \quad D = 2.582$$

$$a_{\text{th}} = 2.582 \quad F = 9.708$$

Then, we write the balanced reaction equation as



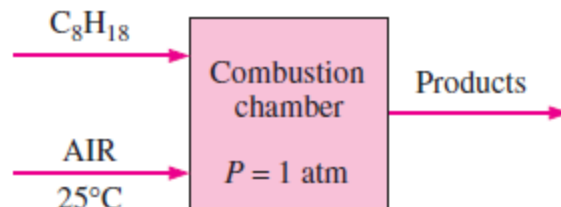
The air-fuel ratio is determined by taking the ratio of the mass of the air to the mass of the fuel,

$$\begin{aligned} \text{AF} &= \frac{m_{\text{air}}}{m_{\text{fuel}}} \\ &= \frac{(2.582 \times 4.76 \text{ kmol})(29 \text{ kg/kmol})}{(0.4182 \text{ kmol})(12 + 4 \times 1) \text{ kg/kmol} + (0.5818 \text{ kmol})(2 \times 12 + 6 \times 1 + 16) \text{ kg/kmol}} \\ &= 10.64 \text{ kg air/kg fuel} \end{aligned}$$

Then, the required flow rate of air becomes

$$\dot{m}_{\text{air}} = \text{AF} \dot{m}_{\text{fuel}} = (10.64)(31 \text{ kg/s}) = \mathbf{330 \text{ kg/s}}$$

15-22 Octane (C_8H_{18}) is burned with 250 percent theoretical air, which enters the combustion chamber at $25^\circ C$. Assuming complete combustion and a total pressure of 1 atm, determine (a) the air–fuel ratio and (b) the dew-point temperature of the products.

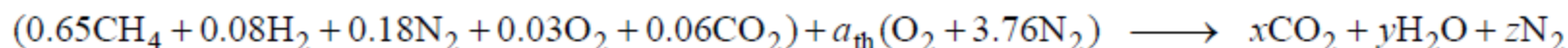


15-23 Gasoline (assumed C_8H_{18}) is burned steadily with air in a jet engine. If the air–fuel ratio is 18 kg air/kg fuel, determine the percentage of theoretical air used during this process.

15-27 A certain natural gas has the following volumetric analysis: 65 percent CH_4 , 8 percent H_2 , 18 percent N_2 , 3 percent O_2 , and 6 percent CO_2 . This gas is now burned completely with the stoichiometric amount of dry air. What is the air–fuel ratio for this combustion process?

15–28 Repeat Prob. 15–27 by replacing the dry air by moist air that enters the combustion chamber at 25°C, 1 atm, and 85 percent relative humidity.

Analysis The fuel is burned completely with the stoichiometric amount of air, and thus the products will contain only H_2O , CO_2 and N_2 , but no free O_2 . The moisture in the air does not react with anything; it simply shows up as additional H_2O in the products. Therefore, we can simply balance the combustion equation using dry air, and then add the moisture to both sides of the equation. Considering 1 kmol of fuel, the combustion equation can be written as



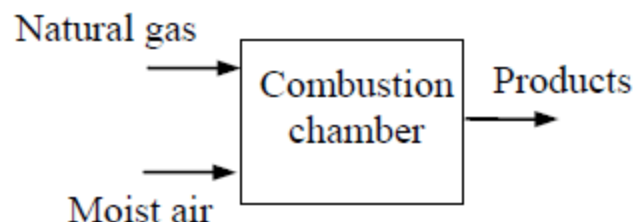
The unknown coefficients in the above equation are determined from mass balances,

$$\text{C} : 0.65 + 0.06 = x \longrightarrow x = 0.71$$

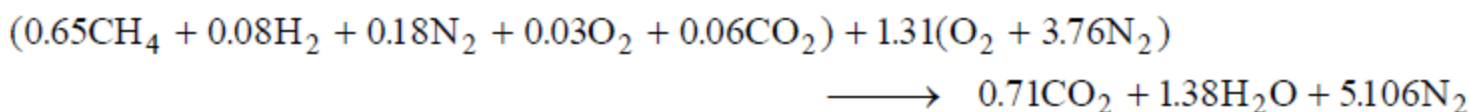
$$\text{H} : 0.65 \times 4 + 0.08 \times 2 = 2y \longrightarrow y = 1.38$$

$$\text{O}_2 : 0.03 + 0.06 + a_{\text{th}} = x + y/2 \longrightarrow a_{\text{th}} = 1.31$$

$$\text{N}_2 : 0.18 + 3.76a_{\text{th}} = z \longrightarrow z = 5.106$$



Thus,



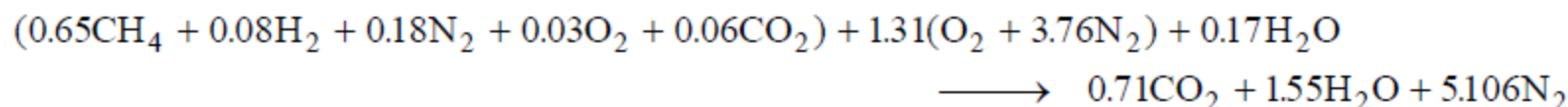
Next we determine the amount of moisture that accompanies $4.76a_{\text{th}} = (4.76)(1.31) = 6.24$ kmol of dry air. The partial pressure of the moisture in the air is

$$P_{v,\text{in}} = \phi_{\text{air}} P_{\text{sat}@25^\circ\text{C}} = (0.85)(3.1698 \text{ kPa}) = 2.694 \text{ kPa}$$

Assuming ideal gas behavior, the number of moles of the moisture in the air ($N_{v, in}$) is determined to be

$$N_{v, in} = \left(\frac{P_{v, in}}{P_{total}} \right) N_{total} = \left(\frac{2.694 \text{ kPa}}{101.325 \text{ kPa}} \right) (6.24 + N_{v, in}) \longrightarrow N_{v, air} = 0.17 \text{ kmol}$$

The balanced combustion equation is obtained by substituting the coefficients determined earlier and adding 0.17 kmol of H_2O to both sides of the equation,



The air-fuel ratio for the this reaction is determined by taking the ratio of the mass of the air to the mass of the fuel,

$$m_{air} = (1.31 \times 4.76 \text{ kmol})(29 \text{ kg/kmol}) + (0.17 \text{ kmol} \times 18 \text{ kg/kmol}) = 183.9 \text{ kg}$$

$$m_{fuel} = (0.65 \times 16 + 0.08 \times 2 + 0.18 \times 28 + 0.03 \times 32 + 0.06 \times 44) \text{ kg} = 19.2 \text{ kg}$$

and

$$AF_{th} = \frac{m_{air, th}}{m_{fuel}} = \frac{183.9 \text{ kg}}{19.2 \text{ kg}} = \mathbf{9.58 \text{ kg air/kg fuel}}$$

15-29 A gaseous fuel with a volumetric analysis of 60 percent CH_4 , 30 percent H_2 , and 10 percent N_2 is burned to completion with 130 percent theoretical air. Determine (a) the air–fuel ratio and (b) the fraction of water vapor that would condense if the product gases were cooled to 20°C at 1 atm.

Answers: (a) 18.6 kg air/kg fuel, (b) 88 percent

15-33 Carbon (C) is burned with dry air. The volumetric analysis of the products is 10.06 percent CO_2 , 0.42 percent CO, 10.69 percent O_2 , and 78.83 percent N_2 . Determine (a) the air–fuel ratio and (b) the percentage of theoretical air used.

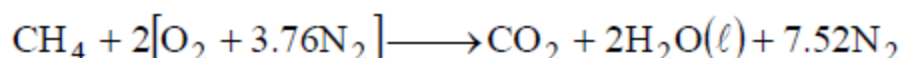
15-34 Methane (CH_4) is burned with dry air. The volumetric analysis of the products on a dry basis is 5.20 percent CO_2 , 0.33 percent CO, 11.24 percent O_2 , and 83.23 percent N_2 . Determine (a) the air–fuel ratio and (b) the percentage of theoretical air used. *Answers: (a) 34.5 kg air/kg fuel, (b) 200 percent*

Enthalpy of Formation and Enthalpy of Combustion

15-42 Determine the enthalpy of combustion of methane (CH_4) at 25°C and 1 atm, using the enthalpy of formation data from Table A-26. Assume that the water in the products is in the liquid form. Compare your result to the value listed in Table A-27. *Answer: -890,330 kJ/kmol*

Assumptions The water in the products is in the liquid phase.

Analysis The stoichiometric equation for this reaction is



Both the reactants and the products are at the standard reference state of 25°C and 1 atm. Also, N_2 and O_2 are stable elements, and thus their enthalpy of formation is zero. Then the enthalpy of combustion of CH_4 becomes

$$h_C = H_P - H_R = \sum N_P \bar{h}_{f,P}^\circ - \sum N_R \bar{h}_{f,R}^\circ = (N \bar{h}_f^\circ)_{\text{CO}_2} + (N \bar{h}_f^\circ)_{\text{H}_2\text{O}} - (N \bar{h}_f^\circ)_{\text{CH}_4}$$

Using $\bar{h}_f^\circ$ values from Table A-26,

$$\begin{aligned} h_C &= (1 \text{ kmol})(-393,520 \text{ kJ/kmol}) + (2 \text{ kmol})(-285,830 \text{ kJ/kmol}) \\ &\quad - (1 \text{ kmol})(-74,850 \text{ kJ/kmol}) \\ &= -890,330 \text{ kJ (per kmol CH}_4) \end{aligned}$$

The listed value in Table A-27 is -890,868 kJ/kmol, which is almost identical to the calculated value. Since the water in the products is assumed to be in the liquid phase, this h_C value corresponds to the higher heating value of CH_4 .

15-44 Repeat Prob. 15-42 for gaseous ethane (C_2H_6).

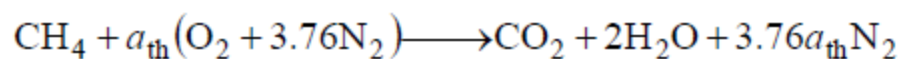
15-45 Repeat Prob. 15-42 for liquid octane (C_8H_{18}).

First-Law Analysis of Reacting Systems

15–49 Methane (CH_4) is burned completely with the stoichiometric amount of air during a steady-flow combustion process. If both the reactants and the products are maintained at 25°C and 1 atm and the water in the products exists in the liquid form, determine the heat transfer from the combustion chamber during this process. What would your answer be if combustion were achieved with 100 percent excess air?

Answer: 890,330 kJ/kmol

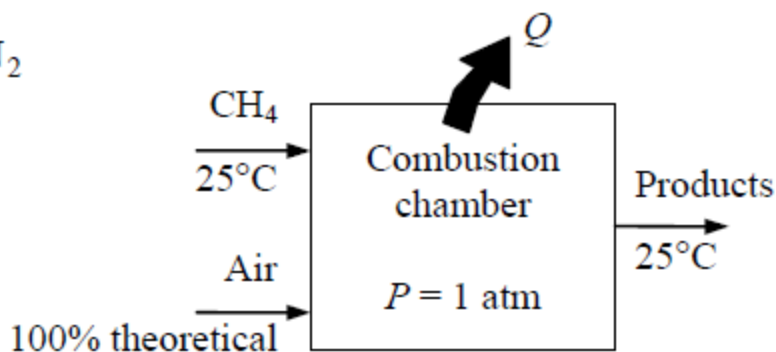
Analysis The fuel is burned completely with the stoichiometric amount of air, and thus the products will contain only H_2O , CO_2 and N_2 , but no free O_2 . Considering 1 kmol of fuel, the theoretical combustion equation can be written as



where a_{th} is determined from the O_2 balance,

$$a_{\text{th}} = 1 + 1 = 2$$

Substituting,



The heat transfer for this combustion process is determined from the energy balance $E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$ applied on the combustion chamber with $W = 0$. It reduces to

$$-Q_{\text{out}} = \sum N_P (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_P - \sum N_R (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_R = \sum N_P \bar{h}_{f,P}^\circ - \sum N_R \bar{h}_{f,R}^\circ$$

since both the reactants and the products are at 25°C and both the air and the combustion gases can be treated as ideal gases. From the tables,

Substance	$\bar{h}_f^\circ$ kJ/kmol
CH ₄	-74,850
O ₂	0
N ₂	0
H ₂ O (ℓ)	-285,830
CO ₂	-393,520

Thus,

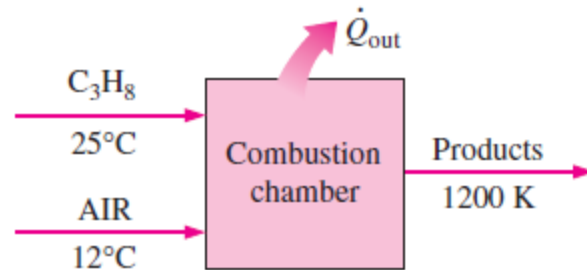
$$-Q_{\text{out}} = (1)(-393,520) + (2)(-285,830) + 0 - (1)(-74,850) - 0 - 0 = -890,330 \text{ kJ / kmol CH}_4$$

or

$$Q_{\text{out}} = 890,330 \text{ kJ / kmol CH}_4$$

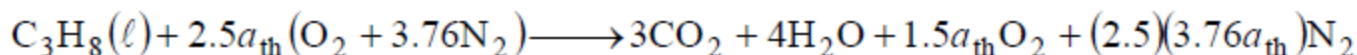
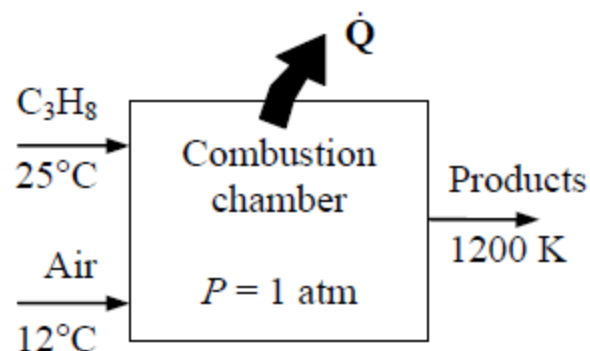
If combustion is achieved with 100% excess air, the answer would still be the same since it would enter and leave at 25°C, and absorb no energy.

15–51 Liquid propane (C_3H_8) enters a combustion chamber at 25°C at a rate of 1.2 kg/min where it is mixed and burned with 150 percent excess air that enters the combustion chamber at 12°C . If the combustion is complete and the exit temperature of the combustion gases is 1200 K , determine (a) the mass flow rate of air and (b) the rate of heat transfer from the combustion chamber. *Answers: (a) 47.1 kg/min , (b) 5194 kJ/min*



Properties The molar masses of propane and air are 44 kg/kmol and 29 kg/kmol , respectively (Table A-1).

Analysis The fuel is burned completely with excess air, and thus the products will contain only CO_2 , H_2O , N_2 , and some free O_2 . Considering 1 kmol of C_3H_8 , the combustion equation can be written as



where a_{th} is the stoichiometric coefficient and is determined from the O_2 balance,

$$2.5a_{\text{th}} = 3 + 2 + 1.5a_{\text{th}} \longrightarrow a_{\text{th}} = 5$$

Thus,



(a) The air-fuel ratio for this combustion process is

$$\text{AF} = \frac{m_{\text{air}}}{m_{\text{fuel}}} = \frac{(12.5 \times 4.76\text{ kmol})(29\text{ kg/kmol})}{(3\text{ kmol})(12\text{ kg/kmol}) + (4\text{ kmol})(2\text{ kg/kmol})} = 39.22\text{ kg air/kg fuel}$$

Thus, $\dot{m}_{\text{air}} = (\text{AF})(\dot{m}_{\text{fuel}}) = (39.22\text{ kg air/kg fuel})(1.2\text{ kg fuel/min}) = \mathbf{47.1\text{ kg air/min}}$

(b) The heat transfer for this combustion process is determined from the energy balance

$E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$ applied on the combustion chamber with $W = 0$. It reduces to

$$-Q_{\text{out}} = \sum N_P (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_P - \sum N_R (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_R$$

Assuming the air and the combustion products to be ideal gases, we have $h = h(T)$. From the tables,

Substance	$\bar{h}_f^\circ$ kJ/kmol	$\bar{h}_{285 \text{ K}}$ kJ/kmol	$\bar{h}_{298 \text{ K}}$ kJ/kmol	$\bar{h}_{1200 \text{ K}}$ kJ/kmol
C ₃ H ₈ (ℓ)	-118,910	---	---	---
O ₂	0	8296.5	8682	38,447
N ₂	0	8286.5	8669	36,777
H ₂ O (g)	-241,820	---	9904	44,380
CO ₂	-393,520	---	9364	53,848

The $\bar{h}_f^\circ$ of liquid propane is obtained by adding $\bar{h}_{fg}$ of propane at 25°C to $\bar{h}_f^\circ$ of gas propane.

Substituting,

$$\begin{aligned} -Q_{\text{out}} &= (3)(-393,520 + 53,848 - 9364) + (4)(-241,820 + 44,380 - 9904) + (7.5)(0 + 38,447 - 8682) \\ &\quad + (47)(0 + 36,777 - 8669) - (1)(-118,910 + h_{298} - h_{298}) - (12.5)(0 + 8296.5 - 8682) \\ &\quad - (47)(0 + 8286.5 - 8669) \\ &= -190,464 \text{ kJ/kmol C}_3\text{H}_8 \end{aligned}$$

or $Q_{\text{out}} = 190,464 \text{ kJ/kmol C}_3\text{H}_8$

Then the rate of heat transfer for a mass flow rate of 1.2 kg/min for the propane becomes

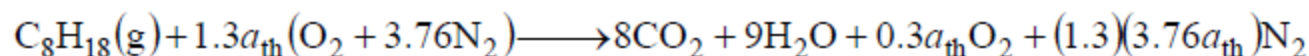
$$\dot{Q}_{\text{out}} = \dot{N}Q_{\text{out}} = \left(\frac{\dot{m}}{N} \right) Q_{\text{out}} = \left(\frac{1.2 \text{ kg/min}}{44 \text{ kg/kmol}} \right) (190,464 \text{ kJ/kmol}) = \mathbf{5194 \text{ kJ/min}}$$

15-55 Benzene gas (C_6H_6) at 25°C is burned during a steady-flow combustion process with 95 percent theoretical air that enters the combustion chamber at 25°C . All the hydrogen in the fuel burns to H_2O , but part of the carbon burns to CO . If the products leave at 1000 K , determine (a) the mole fraction of the CO in the products and (b) the heat transfer from the combustion chamber during this process. *Answers: (a) 2.1 percent, (b) $2,112,800\text{ kJ/kmol C}_6\text{H}_6$*



Octane gas (C_8H_{18}) at 25°C is burned steadily with 30 percent excess air at 25°C , 1 atm, and 60 percent relative humidity. Assuming combustion is complete and the products leave the combustion chamber at 600 K, determine the heat transfer for this process per unit mass of octane.

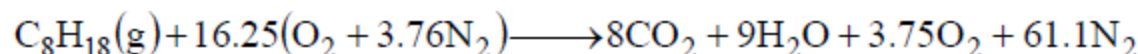
Analysis The fuel is burned completely with the excess air, and thus the products will contain only CO_2 , H_2O , N_2 , and some free O_2 . The moisture in the air does not react with anything; it simply shows up as additional H_2O in the products. Therefore, for simplicity, we will balance the combustion equation using dry air, and then add the moisture to both sides of the equation. Considering 1 kmol of C_8H_{18} , the combustion equation can be written as



where a_{th} is the stoichiometric coefficient for air. It is determined from

$$\text{O}_2 \text{ balance: } 1.3a_{\text{th}} = 8 + 4.5 + 0.3a_{\text{th}} \longrightarrow a_{\text{th}} = 12.5$$

Thus,

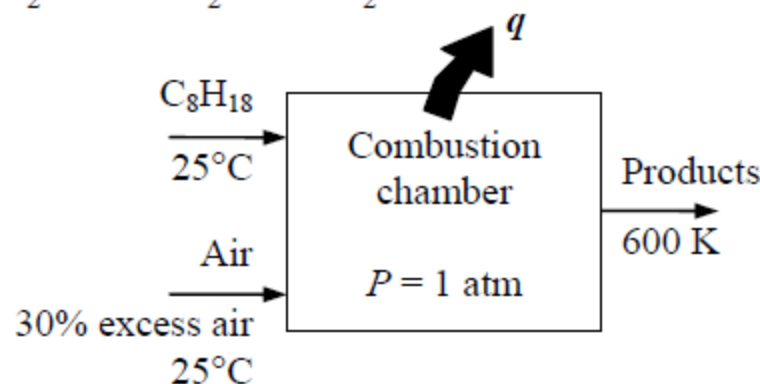


Therefore, $16.25 \times 4.76 = 77.35$ kmol of dry air will be used per kmol of the fuel. The partial pressure of the water vapor present in the incoming air is

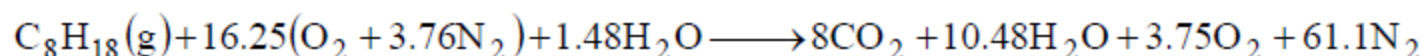
$$P_{v,\text{in}} = \phi_{\text{air}} P_{\text{sat}@25^\circ\text{C}} = (0.60)(3.1698 \text{ kPa}) = 1.902 \text{ kPa}$$

Assuming ideal gas behavior, the number of moles of the moisture that accompanies 77.35 kmol of incoming dry air is determined to be

$$N_{v,\text{in}} = \left(\frac{P_{v,\text{in}}}{P_{\text{total}}} \right) N_{\text{total}} = \left(\frac{1.902 \text{ kPa}}{101.325 \text{ kPa}} \right) (77.35 + N_{v,\text{in}}) \longrightarrow N_{v,\text{in}} = 1.48 \text{ kmol}$$



The balanced combustion equation is obtained by adding 1.48 kmol of H_2O to both sides of the equation,



The heat transfer for this combustion process is determined from the energy balance $E_{in} - E_{out} = \Delta E_{system}$ applied on the combustion chamber with $W = 0$. It reduces to

$$-Q_{out} = \sum N_P (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_P - \sum N_R (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_R = \sum N_P (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_P - \sum N_R \bar{h}_{f,R}^\circ$$

since all of the reactants are at $25^\circ C$. Assuming the air and the combustion products to be ideal gases, we have $h = h(T)$. From the tables,

Substance	$\bar{h}_f^\circ$ kJ/kmol	$\bar{h}_{298\text{ K}}$ kJ/kmol	$\bar{h}_{600\text{ K}}$ kJ/kmol
$C_8H_{18}(g)$	-208,450	---	---
O_2	0	8682	17,929
N_2	0	8669	17,563
$H_2O(g)$	-241,820	9904	20,402
CO_2	-393,520	9364	22,280

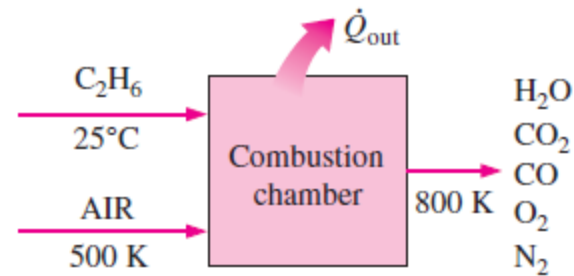
Substituting,

$$\begin{aligned} -Q_{out} &= (8)(-393,520 + 22,280 - 9364) + (10.48)(-241,820 + 20,402 - 9904) \\ &\quad + (3.75)(0 + 17,929 - 8682) + (61.1)(0 + 17,563 - 8669) \\ &\quad - (1)(-208,450) - (1.48)(-241,820) - 0 - 0 \\ &= -4,324,643 \text{ kJ/kmol } C_8H_{18} \end{aligned}$$

Thus 4,324,643 kJ of heat is transferred from the combustion chamber for each kmol (114 kg) of C_8H_{18} . Then the heat transfer per kg of C_8H_{18} becomes

$$q = \frac{Q_{out}}{M} = \frac{4,324,643 \text{ kJ}}{114 \text{ kg}} = 37,935 \text{ kJ/kg } C_8H_{18}$$

15–60 Ethane gas (C_2H_6) at 25°C is burned in a steady-flow combustion chamber at a rate of 5 kg/h with the stoichiometric amount of air, which is preheated to 500 K before entering the combustion chamber. An analysis of the combustion gases reveals that all the hydrogen in the fuel burns to H_2O but only 95 percent of the carbon burns to CO_2 , the remaining 5 percent forming CO. If the products leave the combustion chamber at 800 K, determine the rate of heat transfer from the combustion chamber. *Answer: 200,170 kJ/h*





A constant-volume tank contains a mixture of 120 g of methane (CH_4) gas and 600 g of O_2 at 25°C and 200 kPa. The contents of the tank are now ignited, and the methane gas burns completely. If the final temperature is 1200 K, determine (a) the final pressure in the tank and (b) the heat transfer during this process.

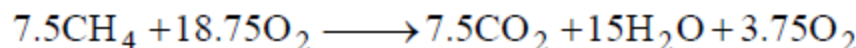
Properties The molar masses of CH_4 and O_2 are 16 kg/kmol and 32 kg/kmol, respectively (Table A-1).

Analysis (a) The combustion is assumed to be complete, and thus all the carbon in the methane burns to CO_2 and all of the hydrogen to H_2O . The number of moles of CH_4 and O_2 in the tank are

$$N_{\text{CH}_4} = \frac{m_{\text{CH}_4}}{M_{\text{CH}_4}} = \frac{0.12 \text{ kg}}{16 \text{ kg/kmol}} = 7.5 \times 10^{-3} \text{ kmol} = 7.5 \text{ mol}$$

$$N_{\text{O}_2} = \frac{m_{\text{O}_2}}{M_{\text{O}_2}} = \frac{0.6 \text{ kg}}{32 \text{ kg/kmol}} = 18.75 \times 10^{-3} \text{ kmol} = 18.75 \text{ mol}$$

Then the combustion equation can be written as



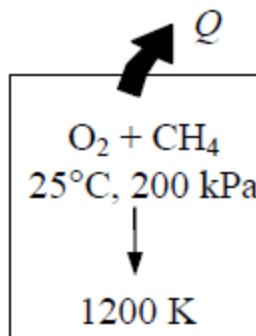
At 1200 K, water exists in the gas phase. Assuming both the reactants and the products to be ideal gases, the final pressure in the tank is determined to be

$$\left. \begin{aligned} P_R V &= N_R R_u T_R \\ P_P V &= N_P R_u T_P \end{aligned} \right\} P_P = P_R \left(\frac{N_P}{N_R} \right) \left(\frac{T_P}{T_R} \right)$$

Substituting,

$$P_P = (200 \text{ kPa}) \left(\frac{26.25 \text{ mol}}{26.25 \text{ mol}} \right) \left(\frac{1200 \text{ K}}{298 \text{ K}} \right) = \mathbf{805 \text{ kPa}}$$

which is relatively low. Therefore, the ideal gas assumption utilized earlier is appropriate.



(b) The heat transfer for this constant volume combustion process is determined from the energy balance $E_{in} - E_{out} = \Delta E_{system}$ applied on the combustion chamber with $W = 0$. It reduces to

$$-Q_{out} = \sum N_P (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ - P\bar{v})_P - \sum N_R (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ - P\bar{v})_R$$

Since both the reactants and products are assumed to be ideal gases, all the internal energy and enthalpies depend on temperature only, and the $P\bar{v}$ terms in this equation can be replaced by $R_u T$. It yields

$$-Q_{out} = \sum N_P (\bar{h}_f^\circ + \bar{h}_{1200\text{ K}} - \bar{h}_{298\text{ K}} - R_u T)_P - \sum N_R (\bar{h}_f^\circ - R_u T)_R$$

since the reactants are at the standard reference temperature of 25°C. From the tables,

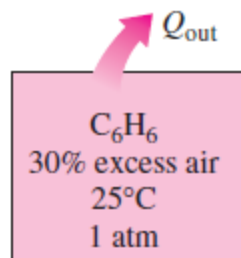
Substance	$\bar{h}_f^\circ$ kJ/kmol	$\bar{h}_{298\text{ K}}$ kJ/kmol	$\bar{h}_{1200\text{ K}}$ kJ/kmol
CH ₄	-74,850	---	---
O ₂	0	8682	38,447
H ₂ O (g)	-241,820	9904	44,380
CO ₂	-393,520	9364	53,848

Thus,

$$\begin{aligned} -Q_{out} &= (7.5)(-393,520 + 53,848 - 9364 - 8.314 \times 1200) \\ &\quad + (15)(-241,820 + 44,380 - 9904 - 8.314 \times 1200) \\ &\quad + (3.75)(0 + 38,447 - 8682 - 8.314 \times 1200) \\ &\quad - (7.5)(-74,850 - 8.314 \times 298) - (18.75)(-8.314 \times 298) \\ &= -5,251,791 \text{ J} = -5252 \text{ kJ} \end{aligned}$$

Thus $Q_{out} = 5252 \text{ kJ}$ of heat is transferred from the combustion chamber as 120 g of CH₄ burned in this combustion chamber.

15-64 A constant-volume tank contains a mixture of 1 kmol of benzene (C_6H_6) gas and 30 percent excess air at $25^\circ C$ and 1 atm. The contents of the tank are now ignited, and all the hydrogen in the fuel burns to H_2O but only 92 percent of the carbon burns to CO_2 , the remaining 8 percent forming CO . If the final temperature in the tank is 1000 K, determine the heat transfer from the combustion chamber during this process.



15-66 To supply heated air to a house, a high-efficiency gas furnace burns gaseous propane (C_3H_8) with a combustion efficiency of 96 percent. Both the fuel and 140 percent theoretical air are supplied to the combustion chamber at $25^\circ C$ and 100 kPa, and the combustion is complete. Because this is a high-efficiency furnace, the product gases are cooled to $25^\circ C$ and 100 kPa before leaving the furnace. To maintain the house at the desired temperature, a heat transfer rate of 31,650 kJ/h is required from the furnace. Determine the volume of water condensed from the product gases per day.

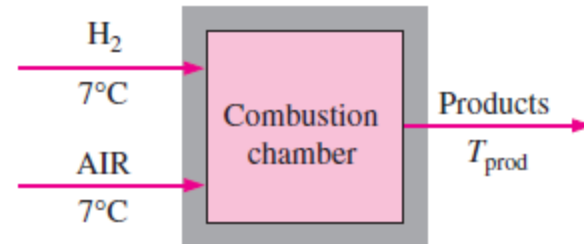
Answer: 8.7 L/day

15–70



Hydrogen (H_2) at 7°C is burned with 20 percent excess air that is also at 7°C during an adiabatic steady-flow combustion process. Assuming complete combustion, determine the exit temperature of the product gases. **Answer: 2251.4 K**

Adiabatic Flame Temperature



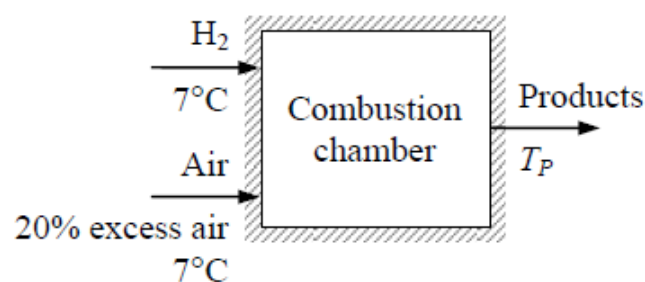
Analysis Adiabatic flame temperature is the temperature at which the products leave the combustion chamber under adiabatic conditions ($Q = 0$) with no work interactions ($W = 0$). Under steady-flow conditions the energy balance $E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$ applied on the combustion chamber reduces to

$$\sum N_P (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_P = \sum N_R (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_R$$

The combustion equation of H_2 with 20% excess air is



From the tables,



Substance	$\bar{h}_f^\circ$ kJ/kmol	$\bar{h}_{280\text{ K}}$ kJ/kmol	$\bar{h}_{298\text{ K}}$ kJ/kmol
H_2	0	7945	8468
O_2	0	8150	8682
N_2	0	8141	8669
$\text{H}_2\text{O} (g)$	-241,820	9296	9904

Thus,

$$\begin{aligned} & (1)(-241,820 + \bar{h}_{\text{H}_2\text{O}} - 9904) + (0.1)(0 + \bar{h}_{\text{O}_2} - 8682) + (2.256)(0 + \bar{h}_{\text{N}_2} - 8669) \\ & = (1)(0 + 7945 - 8468) + (0.6)(0 + 8150 - 8682) + (2.256)(0 + 8141 - 8669) \end{aligned}$$



It yields

$$\bar{h}_{\text{H}_2\text{O}} + 0.1\bar{h}_{\text{O}_2} + 2.256\bar{h}_{\text{N}_2} = 270,116 \text{ kJ}$$

The adiabatic flame temperature is obtained from a trial and error solution. A first guess is obtained by dividing the right-hand side of the equation by the total number of moles, which yields $270,116/(1 + 0.1 + 2.256) = 80,488 \text{ kJ/kmol}$. This enthalpy value corresponds to about 2400 K for N_2 . Noting that the majority of the moles are N_2 , T_P will be close to 2400 K, but somewhat under it because of the higher specific heat of H_2O .

$$\begin{aligned} \text{At 2300 K:} \quad \bar{h}_{\text{H}_2\text{O}} + 0.1\bar{h}_{\text{O}_2} + 2.256\bar{h}_{\text{N}_2} &= (1)(98,199) + (0.1)(79,316) + (2.256)(75,676) \\ &= 276,856 \text{ kJ (Higher than 270,116 kJ)} \end{aligned}$$

$$\begin{aligned} \text{At 2250 K:} \quad \bar{h}_{\text{H}_2\text{O}} + 0.1\bar{h}_{\text{O}_2} + 2.256\bar{h}_{\text{N}_2} &= (1)(95,562) + (0.1)(77,397) + (2.256)(73,856) \\ &= 269,921 \text{ kJ (Lower than 270,116 kJ)} \end{aligned}$$

$$\text{By interpolation,} \quad T_P = \mathbf{2251.4 \text{ K}}$$

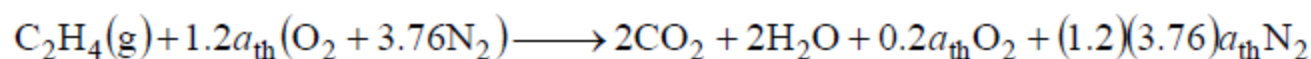
15-74 An adiabatic constant-volume tank contains a mixture of 1 kmol of hydrogen (H_2) gas and the stoichiometric amount of air at 25°C and 1 atm. The contents of the tank are now ignited. Assuming complete combustion, determine the final temperature in the tank.

Entropy Change and Second-Law Analysis of Reacting Systems

15-80 One kmol of H_2 at 25°C and 1 atm is burned steadily with 0.5 kmol of O_2 at the same state. The H_2O formed during the process is then brought to 25°C and 1 atm, the conditions of the surroundings. Assuming combustion is complete, determine the reversible work and exergy destruction for this process.

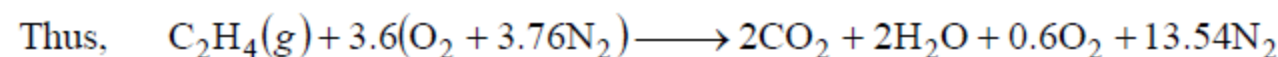
15–81 Ethylene (C_2H_4) gas enters an adiabatic combustion chamber at 25°C and 1 atm and is burned with 20 percent excess air that enters at 25°C and 1 atm. The combustion is complete, and the products leave the combustion chamber at 1 atm pressure. Assuming $T_0 = 25^\circ\text{C}$, determine (a) the temperature of the products, (b) the entropy generation, and (c) the exergy destruction. *Answers: (a) 2269.6 K, (b) 1311.3 kJ/kmol · K, (c) 390,760 kJ/kmol*

Analysis (a) The fuel is burned completely with the excess air, and thus the products will contain only CO_2 , H_2O , N_2 , and some free O_2 . Considering 1 kmol of C_2H_4 , the combustion equation can be written as



where a_{th} is the stoichiometric coefficient and is determined from the O_2 balance,

$$1.2a_{\text{th}} = 2 + 1 + 0.2a_{\text{th}} \longrightarrow a_{\text{th}} = 3$$



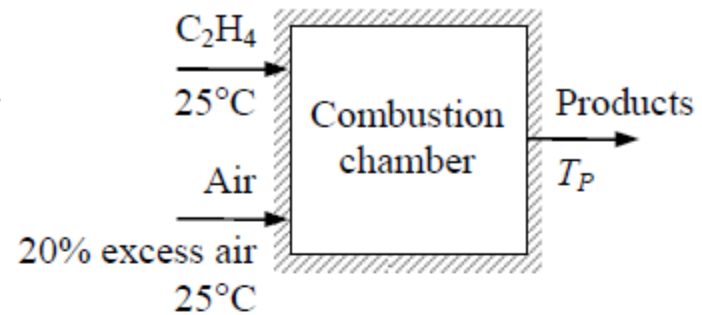
Under steady-flow conditions, the exit temperature of the product gases can be determined from the steady-flow energy equation, which reduces to

$$\sum N_P(\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_P = \sum N_R\bar{h}_{f,R}^\circ = (N\bar{h}_f^\circ)_{\text{C}_2\text{H}_4}$$

since all the reactants are at the standard reference state, and for O_2 and N_2 . From the tables,



Substance	$\bar{h}_f^\circ$ kJ/kmol	$\bar{h}_{298\text{ K}}$ kJ/kmol
$\text{C}_2\text{H}_4 (g)$	52,280	---
O_2	0	8682
N_2	0	8669
$\text{H}_2\text{O} (g)$	-241,820	9904
CO_2	-393,520	9364



Substituting,

$$(2)(-393,520 + \bar{h}_{\text{CO}_2} - 9364) + (2)(-241,820 + \bar{h}_{\text{H}_2\text{O}} - 9904) + (0.6)(0 + \bar{h}_{\text{O}_2} - 8682) + (13.54)(0 + \bar{h}_{\text{N}_2} - 8669) = (1)(52,280)$$

or,

$$2\bar{h}_{\text{CO}_2} + 2\bar{h}_{\text{H}_2\text{O}} + 0.6\bar{h}_{\text{O}_2} + 13.54\bar{h}_{\text{N}_2} = 1,484,083 \text{ kJ}$$

By trial and error,

$$T_P = 2269.6 \text{ K}$$

(b) The entropy generation during this adiabatic process is determined from

$$S_{\text{gen}} = S_P - S_R = \sum N_P \bar{s}_P - \sum N_R \bar{s}_R$$

The C_2H_4 is at 25°C and 1 atm, and thus its absolute entropy is $219.83 \text{ kJ/kmol}\cdot\text{K}$ (Table A-26). The entropy values listed in the ideal gas tables are for 1 atm pressure. Both the air and the product gases are at a total pressure of 1 atm, but the entropies are to be calculated at the partial pressure of the components which is equal to $P_i = y_i P_{\text{total}}$, where y_i is the mole fraction of component i . Also,

$$S_i = N_i \bar{s}_i(T, P_i) = N_i (\bar{s}_i^\circ(T, P_0) - R_u \ln(y_i P_m))$$

The entropy calculations can be presented in tabular form as

	N_i	y_i	$\bar{s}_i^\circ(T, 1\text{atm})$	$R_u \ln(y_i P_m)$	$N_i \bar{s}_i$
C ₂ H ₄	1	1.00	219.83	---	219.83
O ₂	3.6	0.21	205.14	-12.98	784.87
N ₂	13.54	0.79	191.61	-1.96	2620.94
$S_R = 3625.64 \text{ kJ/K}$					
CO ₂	2	0.1103	316.881	-18.329	670.42
H ₂ O	2	0.1103	271.134	-18.329	578.93
O ₂	0.6	0.0331	273.467	-28.336	181.08
N ₂	13.54	0.7464	256.541	-2.432	3506.49
$S_P = 4936.92 \text{ kJ/K}$					

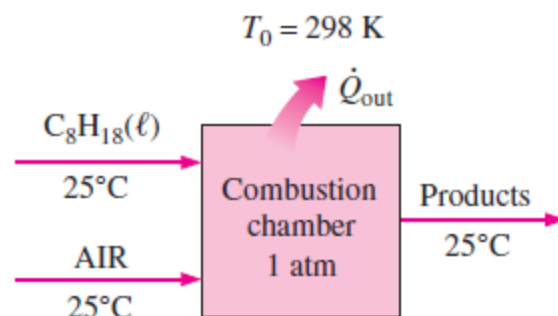
Thus,

$$S_{\text{gen}} = S_P - S_R = 4936.92 - 3625.64 = \mathbf{1311.28 \text{ kJ/kmol} \cdot \text{K}}$$

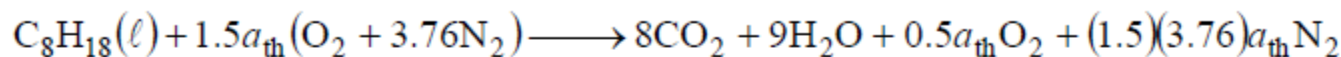
and

$$(c) \quad X_{\text{destroyed}} = T_0 S_{\text{gen}} = (298 \text{ K})(1311.28 \text{ kJ/kmol} \cdot \text{K C}_2\text{H}_4) = \mathbf{390,760 \text{ kJ (per kmol C}_2\text{H}_4)}$$

15–82 Liquid octane (C_8H_{18}) enters a steady-flow combustion chamber at 25°C and 1 atm at a rate of 0.25 kg/min. It is burned with 50 percent excess air that also enters at 25°C and 1 atm. After combustion, the products are allowed to cool to 25°C . Assuming complete combustion and that all the H_2O in the products is in liquid form, determine (a) the heat transfer rate from the combustion chamber, (b) the entropy generation rate, and (c) the exergy destruction rate. Assume that $T_0 = 298 \text{ K}$ and the products leave the combustion chamber at 1 atm pressure.



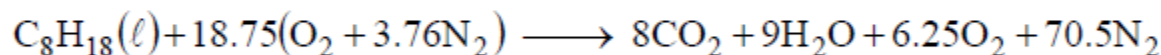
Analysis (a) The fuel is burned completely with the excess air, and thus the products will contain only CO_2 , H_2O , N_2 , and some free O_2 . Considering 1 kmol C_8H_{18} , the combustion equation can be written as



where a_{th} is the stoichiometric coefficient and is determined from the O_2 balance,

$$1.5a_{\text{th}} = 8 + 4.5 + 0.5a_{\text{th}} \longrightarrow a_{\text{th}} = 12.5$$

Thus,

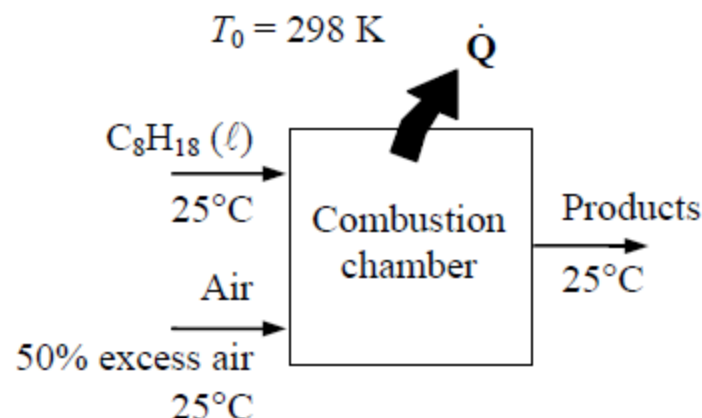


Under steady-flow conditions the energy balance $E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$ applied on the combustion chamber with $W = 0$ reduces to

$$-Q_{\text{out}} = \sum N_P (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_P - \sum N_R (\bar{h}_f^\circ + \bar{h} - \bar{h}^\circ)_R = \sum N_P \bar{h}_{f,P} - \sum N_R \bar{h}_{f,R}$$

since all of the reactants are at 25°C . Assuming the air and the combustion products to be ideal gases, we have $h = h(T)$. From the tables,

Substance	$\bar{h}_f^\circ$ kJ/kmol
$C_8H_{18} (\ell)$	-249,950
O_2	0
N_2	0
$H_2O (l)$	-285,830
CO_2	-393,520



Substituting,

$$-Q_{\text{out}} = (8)(-393,520) + (9)(-285,830) + 0 + 0 - (1)(-249,950) - 0 - 0 = -5,470,680 \text{ kJ/kmol of } C_8H_{18}$$

or

$$Q_{\text{out}} = 5,470,680 \text{ kJ/kmol of } C_8H_{18}$$

The C_8H_{18} is burned at a rate of 0.25 kg/min or

$$\dot{N} = \frac{\dot{m}}{M} = \frac{0.25 \text{ kg/min}}{[(8)(12) + (18)(1)] \text{ kg/kmol}} = 2.193 \times 10^{-3} \text{ kmol/min}$$

Thus,

$$\dot{Q}_{\text{out}} = \dot{N}Q_{\text{out}} = (2.193 \times 10^{-3} \text{ kmol/min})(5,470,680 \text{ kJ/kmol}) = \mathbf{11,997 \text{ kJ/min}}$$

The heat transfer for this process is also equivalent to the enthalpy of combustion of liquid C_8H_{18} , which could easily be determined from Table A-27 to be $\bar{h}_C = 5,470,740 \text{ kJ/kmol } C_8H_{18}$.

(b) The entropy generation during this process is determined from

$$S_{\text{gen}} = S_P - S_R + \frac{Q_{\text{out}}}{T_{\text{surr}}} \longrightarrow S_{\text{gen}} = \sum N_P \bar{s}_P - \sum N_R \bar{s}_R + \frac{Q_{\text{out}}}{T_{\text{surr}}}$$

The C_8H_{18} is at 25°C and 1 atm, and thus its absolute entropy is $\bar{s}_{\text{C}_8\text{H}_{18}} = 360.79 \text{ kJ/kmol}\cdot\text{K}$ (Table A-26).

The entropy values listed in the ideal gas tables are for 1 atm pressure. Both the air and the product gases are at a total pressure of 1 atm, but the entropies are to be calculated at the partial pressure of the components which is equal to $P_i = y_i P_{\text{total}}$, where y_i is the mole fraction of component i . Also,

$$S_i = N_i \bar{s}_i(T, P_i) = N_i \left(\bar{s}_i^\circ(T, P_0) - R_u \ln(y_i P_m) \right)$$

The entropy calculations can be presented in tabular form as

	N_i	y_i	$\bar{s}_i^\circ(\text{T}, 1\text{atm})$	$R_u \ln(y_i P_m)$	$N_i \bar{s}_i$
C_8H_{18}	1	1.00	360.79	---	360.79
O_2	18.75	0.21	205.14	-12.98	4089.75
N_2	70.50	0.79	191.61	-1.96	13646.69
$S_R = 18,097.23 \text{ kJ/K}$					
CO_2	8	0.0944	213.80	-19.62	1867.3
$\text{H}_2\text{O} (\ell)$	9	---	69.92	---	629.3
O_2	6.25	0.0737	205.04	-21.68	1417.6
N_2	70.50	0.8319	191.61	-1.53	13,616.3
$S_P = 17,531 \text{ kJ/K}$					

Thus,

$$S_{\text{gen}} = S_P - S_R + \frac{Q_{\text{surr}}}{T_{\text{surr}}} = 17,531 - 18,097 + \frac{5,470,523 \text{ kJ}}{298 \text{ K}} = 17,798 \text{ kJ/kmol}\cdot\text{K}$$

and

$$\dot{S}_{\text{gen}} = \dot{N} S_{\text{gen}} = (2.193 \times 10^{-3} \text{ kmol/min})(17,798 \text{ kJ/kmol}\cdot\text{K}) = 39.03 \text{ kJ/min}\cdot\text{K}$$

(c) The exergy destruction rate associated with this process is determined from

$$\dot{X}_{\text{destroyed}} = T_0 \dot{S}_{\text{gen}} = (298 \text{ K})(39.03 \text{ kJ/min}\cdot\text{K}) = 11,632 \text{ kJ/min} = 193.9 \text{ kW}$$

15–86

Liquid propane (C_3H_8) enters a steady-flow combustion chamber at 25°C and 1 atm at a rate of 0.4 kg/min where it is mixed and burned with 150 percent excess air that enters the combustion chamber at 12°C . If the combustion products leave at 1200 K and 1 atm, determine (a) the mass flow rate of air, (b) the rate of heat transfer from the combustion chamber, and (c) the rate of entropy generation during this process. Assume $T_0 = 25^\circ\text{C}$.

Answers: (a) 15.7 kg/min, (b) 1732 kJ/min, (c) 34.2 kJ/min · K

Chapter 15

End of the Problems